

SHOCK

By: Mamo Feyissa (BPharm, MSc.)

Assistant professor of Clinical Pharmacy

Department of Pharmacology and Clinical Pharmacy

School of Pharmacy, College of Health Sciences

Addis Ababa University

Definition

- **Shock** involves a group of syndromes that cause acute, generalized circulatory failure associated with inadequate tissue and organ perfusion.
- Shock is characterized by
 - **SBP <90 mm Hg** (or reduction of at least 40 mm Hg from baseline) or
 - MAP<70 mmHg with tachycardia and organ perfusion abnormalities.

- ***Shock** refers to conditions manifested by **hemodynamic alterations** (e.g., hypotension, tachycardia, low cardiac output [CO], and oliguria) caused by:*
 - Intravascular volume deficit (**hypovolemic shock**),
 - Myocardial pump failure (**cardiogenic shock**), or
 - Physical obstruction in the flow of blood (**Obstructive Shock**),
 - Peripheral vasodilation [**distributive shock**] cause by septic, anaphylactic, or neurogenic shock
- Patients may have components of more than one shock syndrome upon presentation

Pathophysiology

- Shock results in failure of the circulatory system to deliver sufficient oxygen (O₂) to tissues resulting → **anaerobic** → ↑ **lactate** conc.
- As the syndrome evolves and compensatory mechanisms are overwhelmed, it becomes increasingly difficult to determine the subtypes because
 - the features of **advanced shock are the same for all.**

Types of chock

1. Hypovolemic shock

- Hemorrhagic
 - GI bleeding, trauma
- Nonhemorrhagic
 - Dehydration, over use of diuretics, sequestration (ascites)

2. Cardiogenic shock

- Nonmechanical causes (e.g., MI)

- Mechanical cause
 - Mitral or aortic insufficiency
 - Rupture of free wall or septum

3. Distributive shock

- Septic shock, anaphylaxis, drug induced

4. Obstructive Shock

Hypovolemic shock

- **Hypovolemic shock:** acute intravascular volume deficiency due to **external losses** or **internal redistribution**.
- ***Inadequate venous return resulting*** insufficient cardiac preload and decreased stroke volume
- This type of shock can be precipitated by
 - **Hemorrhage, burns, trauma, surgery**, intestinal obstruction, and dehydration from considerable insensible fluid loss,
 - **Overaggressive loop-diuretic** administration, and **severe vomiting or diarrhea**.

- **Cardiogenic shock** results from loss of pump function because of decreased cardiac contractility (eg, acute MI], acute valvular abnormality, or arrhythmia (eg, VT).
- **Obstructive shock** results from extracardiac obstruction to blood flow into or out of the heart, such as
 - Tension pneumothorax, cardiac tamponade, or pulmonary embolism.

- **Vasodilatory/distributive shock** is characterized by
 - loss of vascular tone; this involves tissue hypoperfusion due to decreased systemic vascular resistance (or hypoperfusion despite normal or elevated CO).
- **Distributive shock** is a subset of vasodilatory shock characterized by **maldistribution of blood flow** in organ microcirculation.
- Most cases of vasodilatory shock result in distributive shock.
- Almost all cases of **septic shock** result in vasodilatory/distributive shock.

Compensatory Mechanism

- **Abrupt decrease** in tissue perfusion followed by **reperfusion** → **“ischemia-reperfusion injury”**
→ ***Inflammatory Cascade***
- Systemic inflammatory response syndrome (SIRS) with release of numerous mediators that interact to cause further injury-

- **Effects:**
 - Capillary obstruction due to edema
 - Oxygen free radical damage to cell membranes
 - Activation of:
 - **Cellular components:** WBCs, platelets
 - **Humoral components:** Procoagulants, anticoagulants, complement, kinins
 - Release of other inflammatory mediators
 - Formation of microthrombi

- ***Counter-Regulation and Hemodynamic Support***
- **Anti-inflammatory Pathways:**
 - Vagal nerve → Acetylcholine release → Suppresses cytokine production by macrophages
- **Neurohormonal Activation:**
 - **Renin–Angiotensin–Aldosterone System (RAAS):**
 - Angiotensin II → Vasoconstriction
 - Aldosterone → Sodium & water retention
 - **Hypothalamic–Pituitary–Adrenal Axis:**
 - Vasopressin & Cortisol release
 - **Cortisol & Catecholamines:** Inhibit proinflammatory cytokine production
- **Outcome:** Maintain BP, restore perfusion, limit tissue damage

- **Compensatory vasoconstriction** redistributes blood away from the skin, skeletal muscles, kidneys, and GIT **toward vital organs** (e.g., heart, brain).
- **Severe metabolic lactic acidosis** often develops secondary to **tissue ischemia** → localized vasodilation, which further exacerbates the impaired cardiovascular state.

Clinical Presentation of Shock

- **General Symptoms:**
 - Dizziness, lightheadedness, confusion
 - Oliguria (low urine output)
- **Shock-specific Symptoms:**
 - Depend on cause (e.g., fever, malaise in pneumonia; chest pain in AMI)
- **Vital Signs:**
 - Tachycardia (>120 bpm)
 - Tachypnea (>30 breaths/min)
 - Hypotension (SBP <90 mm Hg)
 - Temperature:
 - Normal or low in non-infectious shock (36–37°C / 96.8–98.6°F)
 - Elevated in infection (>38.3°C / 101°F)
- **Physical Exam:**
 - Neurologic: confusion, obtundation
 - Skin: warm (vasodilation) or cool/clammy (vasoconstriction)
 - Delayed capillary refill
 - Decreased urine output

Laboratory Findings in Shock

- **Perfusion Markers:**
 - ↑ Blood lactate (>2 mmol/L)
- **Renal Dysfunction:**
 - ↑ BUN, ↑ creatinine
 - ↓ Urine output (<0.5 – 1 mL/kg/hr)
- **Liver Dysfunction:**
 - ↑ Transaminases
- **Cardiac Markers:**
 - ↑ Troponin (in AMI)
- **Hematologic Findings:**
 - ↓ Hgb/Hct(hemorrhage)
 - ↑ WBC count ($>12,000/\text{mm}^3$) in sepsis
 - Thrombocytopenia
- **Coagulation Abnormalities:**
 - ↑ PT/INR (especially in sepsis or hemorrhagic shock)

- Rapid assessment by **cardiac echocardiography** is indicated when the shock etiology is unclear
- Initial monitoring:
 - vital signs, blood lactate conc., urine production, and physical examination, including mental status
- **Advanced hemodynamic monitoring** (eg, central venous catheterization) may be needed.
 - to measure central venous pressure (CVP),
 - obtain venous samples for laboratory testing (including central venous oxygen saturation [ScvO], and
 - Administer drugs (including vasopressors) or fluids directly into the central circulation
- Venous oximetry (ie, venous oxygen saturations)–
Tissue oxygenation (SCVO₂)

Treatment

- **Goal of therapy:**
- To reduce morbidity and mortality, & halt or reverse existing organ dysfunction.
- **Four phases** , each with different treatment goals and strategies:
 - (1) **Salvage**, minimum perfusion pressure and CO to ensure survival and treating the underlying shock etiology
 - (2) **optimization**, ensuring adequate organ perfusion and Do2
 - (3) **stabilization**, to prevent further end-organ dysfunction; and
 - (4) **de-escalation**, recovery with goals of weaning (or stopping) vasoactive medications and fluids.

- **Hospitalization**
- Venous access (peripheral IV lines or central IV lines for fluid resuscitation)
- **The stabilization phase** should include **general supportive care** measures (assessment and management of pain, anxiety/agitation, delirium, immobility, sleep disturbances, nutrition, glycemic control, and VTE).
- If MAP or SBP remains below goal, then **initiate vasopressors** to ensure tissue perfusion.
- Fig.

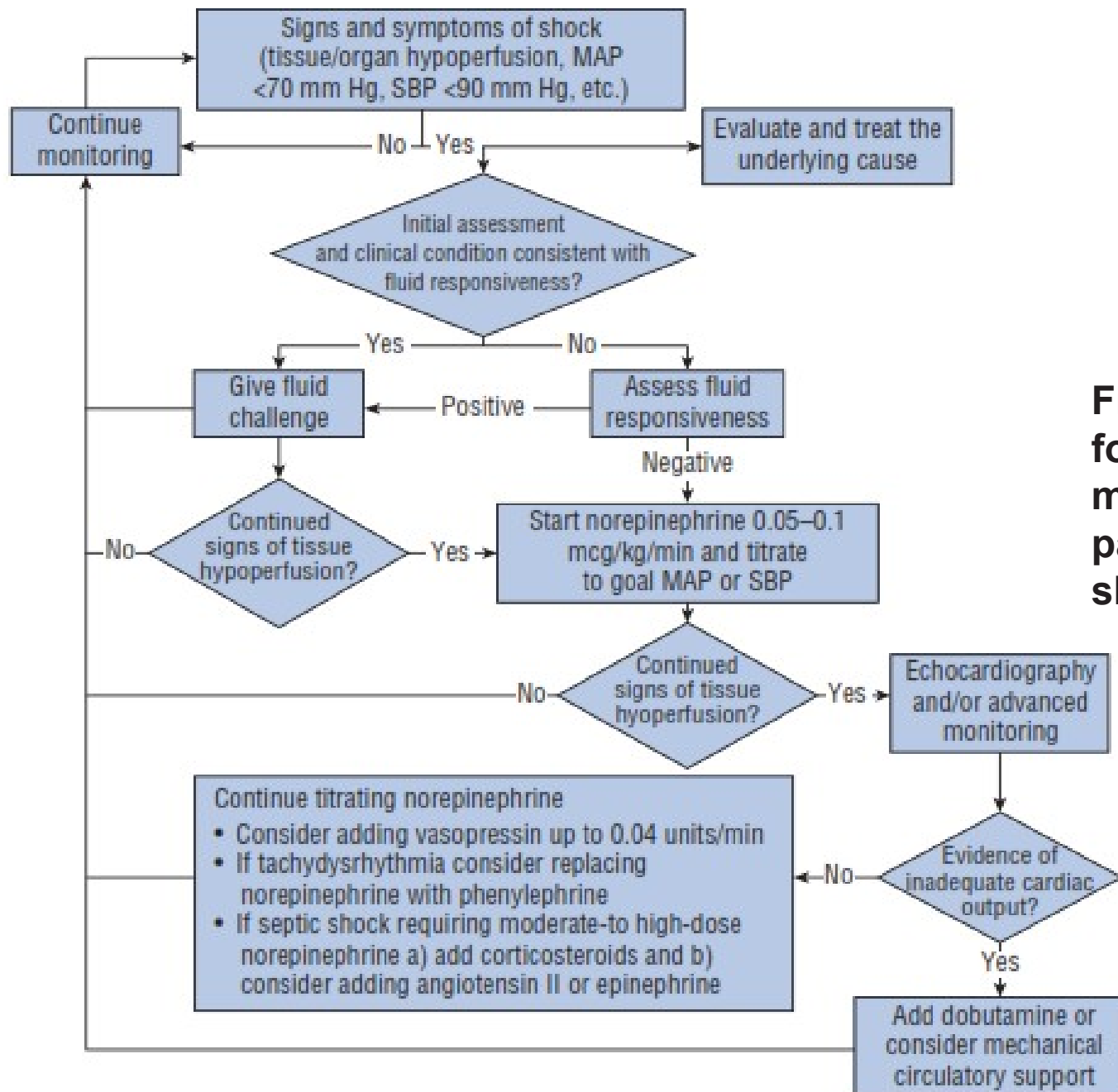


Fig. Algorithm for the management of patients with shock.

NONPHARMACOLOGIC THERAPY

- Basic life support measures; a secure airway with appropriate oxygenation.
- **Hypovolemic shock**,
 - surgery, control of blood loss, blood component transfusion, and prevention of heat loss with hypothermia or cooling measures with heat exposure (eg, heat stroke).
- **Cardiogenic shock** 2^o to AMI
 - Coronary revascularization and
 - **CO augmentation** via mechanical device
- **obstructive shock**
 - Pericardiocentesis or surgical evacuation of fluid, needle decompression and/or chest tube thoracostomy for tension pneumothorax, and surgical or catheter thrombectomy for pulmonary embolism
- **Vasodilatory/distributive shock** 2^o to septic shock
 - infectious source control

FLUID RESUSCITATION FOR HYPOVOLEMIC SHOCK

- Initial fluid resuscitation consists of
 - **Isotonic crystalloid** (0.9% sodium chloride or lactated Ringer's solution),
 - **Colloid** (5% Plasmanate or albumin, 6% hetastarch), or
 - Whole blood.
- **Choice of solution** is based on O₂- carrying capacity, cause of hypovolemic shock, accompanying disease states, degree of fluid loss, and required speed of fluid delivery.

- Crystalloids should be the initial therapy of circulatory insufficiency.
- Crystalloids are preferred over colloids as initial therapy for burn pts because they are less likely to cause interstitial fluid accumulation
- If volume resuscitation is suboptimal following several liters of crystalloid, colloids should be considered

Crystalloids

- Crystalloids consist of electrolytes (Na^+ , Cl^- , K^+) in water solutions, with or without dextrose.
- **Lactated Ringer's** solution may be preferred because it is unlikely to cause the **hyperchloremic metabolic acidosis** seen with infusion of large amounts of normal saline.
- Usually 2 to 4 L of crystalloid normalizes intravascular volume.
 - Approximately 4 L of NS must be infused to replace 1 L of blood loss.

Colloids

- Colloids are larger molecular weight solutions that have been recommended for use in conjunction with or as replacements for crystalloid solutions.
- Include
 - **Albumin**
 - **Hedroxyethyl starch (hetastarch) and**
 - **Dextran solutions**
- Advantages are prolonged IV retention time than crystalloids (remain IV space for hours to days).
- **Costy,**

Blood Products

- **Whole blood** could be used for large volume blood loss, but most institutions use component therapy, with crystalloids or colloids used for plasma expansion.
- **Packed RBC:** are usually indicated in pts with continued deterioration after volume replacement or obvious exsanguination.
- **Fresh frozen plasma** replaces clotting factors.
 - Indicated if there is ongoing hemorrhage in pts with PT or aPTT > 1.5 X normal, severe hepatic disease, or other bleeding disorders

- Risks associated with infusion of blood products include:
 - ◆ Transfusion related reactions, virus transmission (rare),
 - ◆ Hypocalcemia resulting from added citrate,
 - ◆ Elevations in serum K and phosphorus
 - ◆ Increased blood viscosity and
 - ◆ Hypothermia from failure to appropriately warm solutions before administration.

Vasopressors and inotropes

- are required when resuscitation fails to maintain adequate BP (MAP>65 mmHg) and organs and tissues remain hypoperfused
- **Norepinephrine** (0.1–2 mcg/kg/min) is the vasopressor of choice in patients in most shock states.
- **Epinephrine**
- **has combined α - and β -agonist effects. At low doses (0.01–0.05 mcg/kg/min) β -adrenergic effects predominate with an increase in stroke volume and CO.**
 - **low doses** may be used as an inotrope **after cardiac surgery.**
- When higher dosages are used, α -adrenergic effects are predominant and SVR and MAP are increased.
- Epinephrine is an acceptable choice for hemodynamic support of pts with shock.

- **Phenylephrine** is a pure α_1 -agonist and increases BP primarily through vasoconstriction.
- It improves MAP by increasing SVR and stroke index through enhanced venous return to the heart.
- It may also increase contractility and CO (cardiac α_1 -receptors).
- Tachydysrhythmias are infrequent (as a single agent or at higher doses) thus useful alternative in pts who cannot tolerate tachycardia or tachydysrhythmias from E & NE.
- Attractive agent for vasodilatory/distributive shock
- Phenylephrine 0.5–9 mcg/kg/min, used alone or in combination with *dobutamine* or *low dopamine doses*, improves BP and myocardial performance in fluid-resuscitated pts with vasodilatory/distributive shock.

Dopamine

- Is often the initial vasopressor used in septic shock because it increases BP by increasing myocardial contractility and vasoconstriction.
- In pts with septic shock, doses of 5 to 10 mcg/kg/min are initiated to improve MAP
- In septic shock, these doses increase CI by improving ventricular contractility, HR, MAP, and systemic vascular resistance (SVR)

- **Dobutamine** is primarily a selective β_1 -agonist with mild β_2 and vascular α_1 activity, resulting in strong positive inotropic activity without concomitant vasoconstriction.
- Dobutamine produces a larger increase in CO and is less arrhythmogenic than dopamine.
- Clinically, β_2 -induced vasodilation and the increased myocardial contractility with subsequent reflex reduction in sympathetic tone lead to a **decrease in SVR**.
- Dobutamine should be started with doses ranging from 2.5 to 5 mcg/ kg/min.

- **Vasopressin** causes vasoconstrictive effects that, unlike adrenergic receptor agonists, are preserved during hypoxia and severe acidosis.
- Vasopressin may be considered in septic shock pts who are refractory to catecholamine vasopressors despite adequate fluid resuscitation.
- If used, the dose should not exceed 0.01 to 0.04 units/min.

- **Corticosteroids** can be initiated in septic shock when adrenal insufficiency is present or when weaning of vasopressor therapy proves futile.
- A daily dose equivalent to 200 to 300 mg hydrocortisone should be continued for 7 days.

Evaluation of outcome

- The initial monitoring include vital signs, urine output, mental status, and physical examination.
- Cardiovascular and respiratory parameters should be monitored continuously
- Laboratory tests indicated (electrolytes and renal function tests, CBC)
- Successful fluid increase SBP (> 90 mm Hg), CI (>2.2 L/min/m²), and urine output (0.5 to 1 mL/kg/hour)
- MAP > 65 mm Hg should be achieved to ensure adequate cerebral and coronary perfusion pressure.